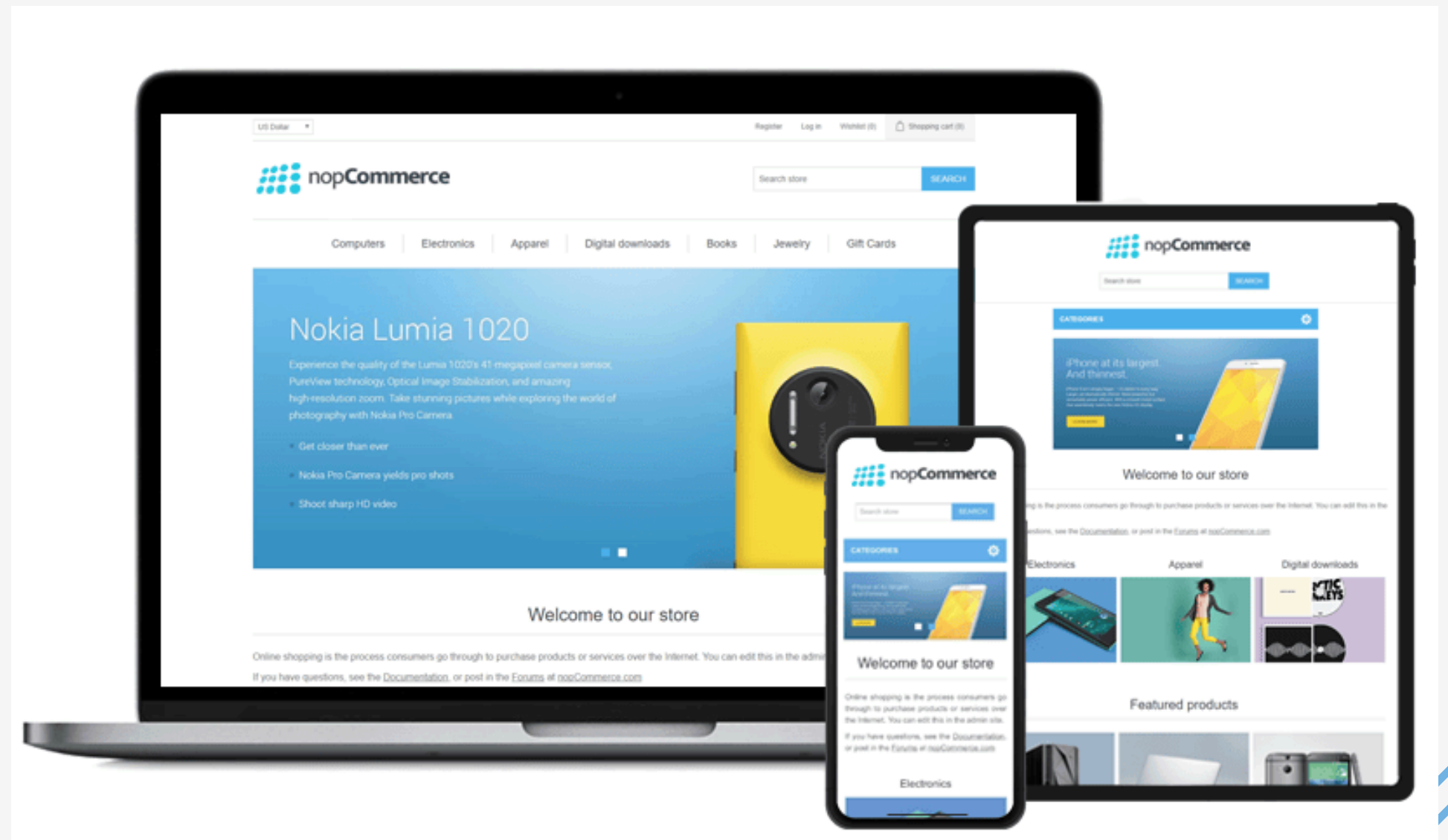


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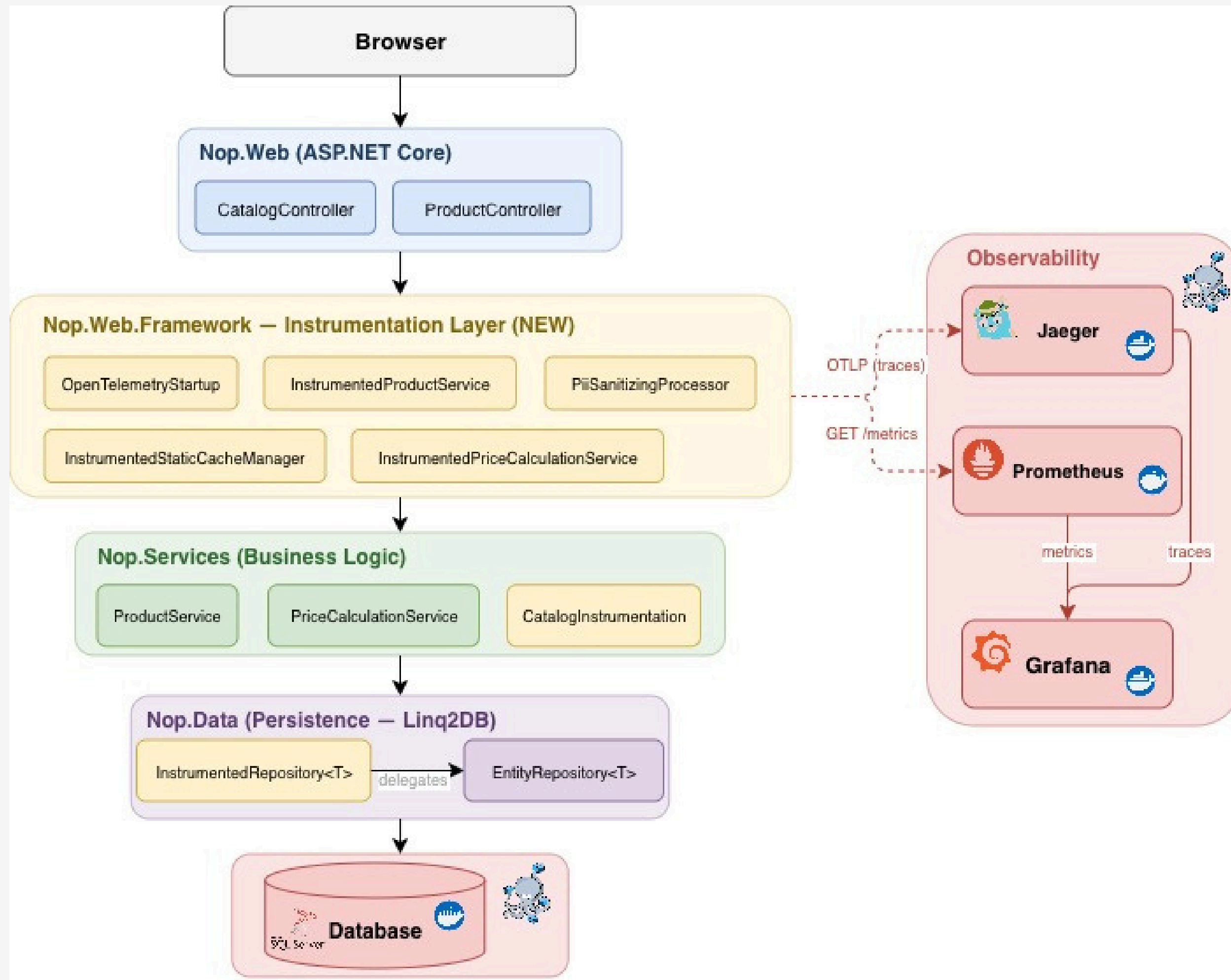
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Assignment 1 - Individual



THE CHALLENGE & CHOSEN FLOW

- **nopCommerce:** open-source e-commerce platform, C# / ASP.NET Core, 9k+ GitHub stars
- **Instrumented flow:** Search → Catalogue → Pricing
 - **Search:** user searches
 - **Catalogue:** clicks a product
 - **Pricing:** price calculation
- **Challenge:** zero existing instrumentation
- **Goal:** end-to-end observability without modifying business logic

ARCHITECTURE



METRICS

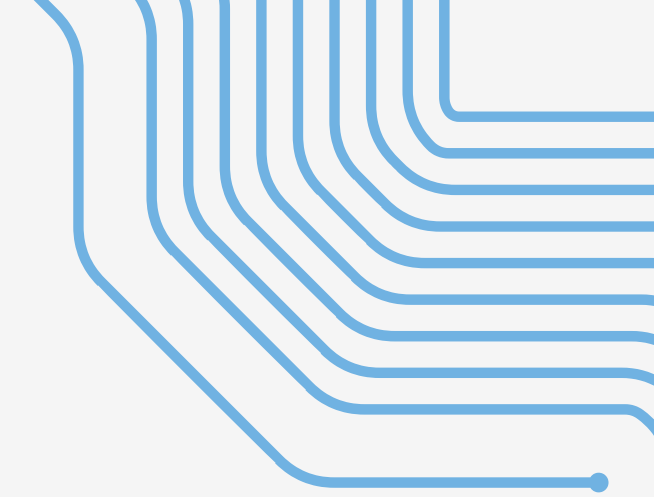
- search.results_count
- product_view_duration_ms
- cache.hits
- cache.misses
- catalog.errors
- database.query_duration_ms



METRICS

4. Distributed Traces (Jaeger)

Search Flow Traces (GET /search/)				Catalogue Core Spans (Product View)			
Trace ID	Trace name	Start time	Duration	Trace ID	Trace name	Start time	Duration
ecfc1f5c3717badb189eac881a84...	nopcommerce-web: GET /search/	2026-03-21 22:54:06.218	48.2 ms	76f0bd29a3f33a37e783ef5d4bef...	nopcommerce-web: GET	2026-03-21 22:54:04.837	55.2 ms
54763ea09d34d4f3555b25ab9a...	nopcommerce-web: GET /search/	2026-03-21 22:54:06.091	27.0 ms	40a307b89572ce2f573097b276b...	nopcommerce-web: GET	2026-03-21 22:54:04.837	50.4 ms
7e30679fd99cd2dd7d06f057a171...	nopcommerce-web: GET /search/	2026-03-21 22:54:06.065	25.1 ms	2012ddcf1ab187ce55e45c27a306...	nopcommerce-web: GET	2026-03-21 22:54:04.764	59.2 ms
6cd0a802e5cf920c209bea7543...	nopcommerce-web: GET /search/	2026-03-21 22:54:06.039	26.1 ms	506dc318fc3b8ea6e7999f0d6f3...	nopcommerce-web: GET	2026-03-21 22:54:04.694	35.2 ms
06ff383a8dd88a76978ff96cdcda...	nopcommerce-web: GET /search/	2026-03-21 22:54:06.011	26.3 ms	35506ad24867ff08f1f38f5da47e...	nopcommerce-web: GET	2026-03-21 22:54:04.688	41.8 ms
69c8ba0b360590eb9bc31e4ee9...	nopcommerce-web: GET /search/	2026-03-21 22:54:06.011	28.1 ms	fb916cf7ff643826444dece8284a...	nopcommerce-web: GET	2026-03-21 22:54:04.683	74.2 ms
c70586b9c4c6e72ec2f92ea1f24...	nopcommerce-web: GET /search/	2026-03-21 22:54:05.992	19.1 ms	878cafd194d8972896aa9d92c88...	nopcommerce-web: GET	2026-03-21 22:54:04.649	39.2 ms
5083a904d1fad2cf588fc6b8f8b6...	nopcommerce-web: GET /search/	2026-03-21 22:54:05.963	28.0 ms	5dda8580875db115aa028115c6d...	nopcommerce-web: GET	2026-03-21 22:54:04.639	64.6 ms
2caff7be63baab25b830520efb2...	nopcommerce-web: GET /search/	2026-03-21 22:54:05.934	29.5 ms	44a39ebd2b4b2c9ce89ba6dacb...	nopcommerce-web: GET	2026-03-21 22:54:04.602	86.4 ms
85f91cb84151b93cb62868a4fedd...	nopcommerce-web: GET /search/	2026-03-21 22:54:05.911	32.4 ms	d04f3c4553a167a8466deccada3...	nopcommerce-web: GET	2026-03-21 22:54:04.475	66.6 ms
Pricing Calculation Traces				Database Layer Traces (Repository.GetAllPaged)			
Trace ID	Trace name	Start time	Duration	Trace ID	Trace name	Start time	Duration
ecfc1f5c3717badb189eac881a84...	nopcommerce-web: GET /search/	2026-03-21 22:54:06.218	48.2 ms	55fb33f61ba489f82ab4804a797...	nopcommerce-web: GET	2026-03-21 22:54:05.930	49.3 ms
b1aa635fe9622863a9083099adf...	nopcommerce-web: GET	2026-03-21 22:54:06.118	37.0 ms	85f91cb84151b93cb62868a4fedd...	nopcommerce-web: GET /search/	2026-03-21 22:54:05.911	32.4 ms
06ff383a8dd88a76978ff96cdcda...	nopcommerce-web: GET /search/	2026-03-21 22:54:06.011	26.3 ms	280242d92e0b31263646688211...	nopcommerce-web: GET /search/	2026-03-21 22:54:05.907	26.9 ms
55fb33f61ba489f82ab4804a797...	nopcommerce-web: GET	2026-03-21 22:54:05.930	49.3 ms	4612252a3fa89f9ac2dc5fba4dac...	nopcommerce-web: GET /search/	2026-03-21 22:54:05.897	32.7 ms
85f91cb84151b93cb62868a4fedd...	nopcommerce-web: GET /search/	2026-03-21 22:54:05.911	32.4 ms	d1c136e2d20d3553ae7e79f8f872...	nopcommerce-web: GET /search/	2026-03-21 22:54:05.870	26.5 ms
573a15f87df2e1faa50442832b66...	nopcommerce-web: GET	2026-03-21 22:54:05.782	34.4 ms	54f4d9b2e656e090dfb7b4138e1...	nopcommerce-web: GET /search/	2026-03-21 22:54:05.841	28.7 ms
cc3d3ce6656316b93d8aa0d902...	nopcommerce-web: GET /search/	2026-03-21 22:54:05.682	55.6 ms	19c0c27d43df61ccee7b8ef5a5ba...	nopcommerce-web: GET /search/	2026-03-21 22:54:05.817	23.7 ms
ac9f90adb739d84144fda6d97e1e...	nopcommerce-web: GET	2026-03-21 22:54:05.606	34.1 ms	4da39623e21dc8e06c56d85fb4a...	nopcommerce-web: GET /search/	2026-03-21 22:54:05.789	28.3 ms
f66630541ff4670eca2845ba72f0...	nopcommerce-web: GET /search/	2026-03-21 22:54:05.580	42.2 ms	573a15f87df2e1faa50442832b66...	nopcommerce-web: GET	2026-03-21 22:54:05.782	34.4 ms
0ae36cf17a3f68b2ef93ba3c8af7...	nopcommerce-web: GET	2026-03-21 22:54:05.500	42.4 ms	5d5ba7f4f6d28b39bb2c6acefb5...	nopcommerce-web: GET /search/	2026-03-21 22:54:05.760	28.7 ms



DEMO

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